Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Applied AI & GenAI Systems · Python / LangGraph / MCP · Evaluation Harness & Real-World-Actuating AI

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SUMMARY

Senior Applied AI Engineer with 9 years of production software ownership and hands-on GenAI system delivery. Built AgentProbe - an open-source evaluation and regression framework with a CI-integrated gate, LLM-as-judge scoring, and multi-trial pass@k harness - the exact tooling needed to keep a non-deterministic, real-world-actuating system reliable. Built LeaseI - an 8-agent LangGraph orchestration platform with prompting, RAG retrieval, tool calling, guardrails, and HITL escalation on regulated decisions - where the core design rule is: LLM reasons, deterministic logic makes every final call. Seven years at Entrata owning systems where a wrong data value has regulated consequences for thousands of residents, with zero compliance violations. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI & GenAI Systems: LangGraph, Agentic AI, LLM Orchestration, Prompting, Retrieval (RAG Pipelines), Tool Calling, HITL, Guardrails, Confidence Scoring, Output Validation, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Regression Frameworks, CI-Integrated Eval Gate, Pass@k Harness, Production AI, Agent Observability, Agent Tracing, MCP (Model Context Protocol), ChromaDB, Vector Database, Semantic Search, OpenAI API, Anthropic API, Groq, Ollama

Languages & Frameworks: Python, TypeScript, JavaScript, Java, SQL, PL/SQL, PHP, R, Bash, HTML/CSS, FastAPI, React.js, Angular, Node.js, Spring Boot, .NET Framework

Data & ML: PostgreSQL, MySQL, Oracle, NoSQL, ETL, Data Modeling, Data Pipelines, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, spaCy, BERT, Hugging Face, Tableau, Power BI

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), RabbitMQ, Redis, Microservices, Distributed Systems, Observability, Production Monitoring, Performance Engineering, Git, Linux, Agile/Scrum, Libsodium

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a production agentic system in LangGraph (parse → plan → retrieve → validate → conditional routing) for external client delivery, end-to-end from design through deployment; made explicit build-vs-buy decisions on orchestration, retrieval, and validation layers with cost and latency tradeoffs documented.
- Implemented RAG over regulatory corpus with guardrails enforcing compliance rules before output -- every response grounded in cited source regulation; non-compliant outputs blocked before export.

Entrata (PropTech / Real Estate SaaS · 100K+ housing units)

Sep 2018 - Aug 2024

Senior Software Engineer

- **Owned compliance-critical data systems where a wrong output has regulated consequences** -- annual HUD income-limit pipeline (few thousand federal rows → 180,000 domain records powering eligibility and rent decisions for 100K+ residents); zero compliance violations across the full ownership period. The discipline: precision first, speed second.
- Built and owned RabbitMQ-based async processing for gross-rent updates across thousands of tenant records -- designed retry logic and idempotency guarantees that survived production incidents and annual rent-schedule changes; never double-processed.
- Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning; owned complex schemas for eligibility, financial, and document workflows across multi-tenant production environments.
- Led AWS S3 document migration -- 100% PHI compliance with Libsodium-encrypted pipelines, zero security incidents; shipped 20+ full-stack production modules (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints; led microservices redesign using Docker (50% scalability improvement, 30% deployment time reduction).
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; eliminated 80% of manual operational workload through Python/Bash automation pipelines.

Entrata India (Db Xento Systems)

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee Software Engineer

- Built property management and compliance modules in Python, Java, and PostgreSQL -- income-limits import (1.6M HUD/Tax Credit records), document management workflows, AWS S3 migration, database architecture redesign.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets -- $R^2 = 0.977$ accuracy; applied Isolation Forest for automated data integrity checks across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: AI Evaluation & Regression Framework

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions)

- Built a production-grade AI evaluation and regression framework: 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate that blocks quality degradation on every commit, LLM-as-judge scoring with multi-trial pass@k harness, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) -- the tooling needed to keep a non-deterministic system reliable before it acts in the real world.
- 15-endpoint FastAPI backend; evaluated LeaseI across 1,240+ compliance test cases with full observability into agent trace and scoring rationale.

LeaseI: Agentic LLM Orchestration Platform with Guardrails

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Streamlit

- Built an 8-agent LangGraph platform with prompting, RAG retrieval, tool calling, conditional routing, HITL escalation, and output validation -- core design rule enforced in code: LLM extracts and reasons; deterministic logic makes every final regulated call. Every answer cited to exact source or flagged as unsupported.
- Evaluated across 1,240 test cases (5 US states + federal): 93.6% precision, 87.9% F1, pass@1 81.7%, pass@3 88.3%; regression gate catches quality regressions before deployment.

AI on FHIR: NLP-Driven Medical Records Query System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · spaCy · BERT · Docker

- Reduced manual clinical retrieval time ~60%; HIPAA-compliant RBAC; NLP-to-FHIR query translation in EN/ES -- a system where wrong data access has regulated consequences, same discipline as compliance-critical AI.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017